

N100 FINANCIAL INTELLIGENCE PLATFORM

ANALYST GUIDE & USER MANUAL

A Comprehensive Guide to Screening, Scoring, Valuation, and API Operations

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Executive Summary

The N100 Financial Intelligence Platform is a production-grade analytics platform designed to monitor, score, and cluster the 92 constituent companies of the Nifty 100 index. Built on top of a robust SQLite database, the platform automates data ingestion (ETL), validates records against 16 core Data Quality (DQ) rules, computes historical financial ratios, ranks companies using a sector-relative composite quality score, categorizes business profiles using unsupervised machine learning (KMeans), and exposes data via both an interactive 8-screen Streamlit dashboard and a 16-endpoint FastAPI REST API.

1. Data Quality Rules & Ingestion Pipeline

To guarantee high data integrity, the ingestion pipeline enforces 16 validation rules. These rules run before any calculation is saved to the SQLite database. Any failures are saved to output/validation_failures.csv.

DQ-01 - Company PK Uniqueness: Ensures 'id' is unique in the companies master table.

DQ-02 - Annual PK Uniqueness: Ensures (company_id, year) is unique across profit & loss, balance sheet, and cashflow tables.

DQ-03 - Foreign Key Integrity: Checks that company_ids in transaction tables exist in companies master.

DQ-04 - Balance Sheet Balance: Verifies assets equal liabilities within a 1% threshold.

DQ-05 - OPM Cross-Check: Ensures computed operating profit margin matches sheet OPM within 1%.

DQ-06 - Positive Sales Check: Flags non-financial companies with zero or negative sales.

DQ-07 - Year Format: Validates that year strings conform to 'YYYY-MM' format.

DQ-08 - Ticker Format: Validates that company tickers are between 2 and 12 uppercase characters.

DQ-09 - Net Cash Flow Check: Verifies net cash flow equals sum of CFO, CFI, and CFF.

DQ-10 - Non-Negative Fixed Assets: Flags negative fixed asset records and coerces to 0.

2. Ratio Engine & Financial Formulas

The `src/analytics/ratios.py` and `src/analytics/cashflow_kpis.py` modules calculate core financial indicators. Key formulas implemented include:

Return on Equity (ROE):

$$\text{Net Profit} / (\text{Equity Capital} + \text{Reserves}) * 100$$

Return on Capital Employed (ROCE):

$$(\text{Profit Before Tax} + \text{Interest}) / (\text{Equity Capital} + \text{Reserves} + \text{Borrowings}) * 100$$

Debt-to-Equity (D/E):

$$\text{Borrowings} / (\text{Equity Capital} + \text{Reserves})$$

Operating Profit Margin (OPM):

$$\text{Operating Profit} / \text{Sales} * 100$$

Interest Coverage Ratio (ICR):

$$(\text{Operating Profit} + \text{Other Income}) / \text{Interest}$$

Free Cash Flow (FCF):

$$\text{Cash from Operations} + \text{Cash from Investing (Capex is negative investing flow)}$$

Book Value Per Share (BVPS):

$$(\text{Equity Capital} + \text{Reserves}) / (\text{Equity Capital} * 10.0)$$

3. Composite Scoring System

The scoring engine (compute_composite_score) calculates a sector-relative quality score (0 to 100) using 10 weighted metrics. This ensures companies are compared fairly against peers in their own sector, avoiding bias (e.g. comparing IT margins to Retail).

Profitability (35%): ROE (15%), ROCE (10%), Net Profit Margin (10%)

Cash Quality (30%): FCF CAGR (15%), CFO/PAT Ratio (10%), FCF Positive Flag (5%)

Growth (20%): 5yr Revenue CAGR (10%), 5yr PAT CAGR (10%)

Leverage (15%): Debt-to-Equity (10%), Interest Coverage Ratio (5%)

The scores are winsorized at P10/P90 percentiles to prevent outliers from distorting the distribution, and scaled so that the top company in each sector scores 100.

4. KMeans Clustering Methodology

Using scikit-learn, the platform clusters all 92 companies into 5 distinct clusters based on their financial features. Prior to clustering, missing values are imputed with sector medians and all features are scaled using StandardScaler.

Input Features used: return_on_equity_pct, debt_to_equity, revenue_cagr_5yr, fcf_cagr_5yr, operating_profit_margin_pct

Cluster Archetypes & Identifiers:

High-Quality Compounders: High ROE, strong operating margins, low leverage, consistent cash generation.

Emerging Growth: High sales and FCF CAGR growth rates, moderate ROE, reinvesting cash flows aggressively.

Defensive Dividend Payers: Low leverage, highly stable interest coverage, steady cash distribution, mature profile.

Value Cyclical: Higher debt-to-equity, volatile margins, moderate ROE, asset-heavy structure.

Distressed or Turnaround: Negative or low growth, depressed ROE, low interest coverage, potential leverage warnings.

5. Streamlit Dashboard User Interface Navigation

The platform offers an interactive 8-screen dashboard accessible via: `streamlit run src/dashboard/app.py`

01 Home Screen: Displays aggregate metrics, sector distribution donut, and top-5 leaderboard.

02 Company Profile: Search box to inspect profile details, 10-yr trends, and dynamic chart comparisons.

03 Screener Screen: Interactive sliders and presets (Quality, Value, Growth, Dividend, Turnaround) with CSV download.

04 Peer Comparison: benchmarking with a gold highlight for the peer group benchmark company.

05 Trend Analysis: YoY line overlays for up to 3 metrics over 10 years.

06 Sector Analysis: Bubble chart (Sales vs ROE vs Cap) and cross-sector median comparisons.

07 Capital Allocation: Treemap of cash flow patterns (Reinvestor, Liquidating, etc.).

08 Annual Reports: Annual report link explorer with live URL status badges.

6. PDF Tearsheets and Sector Reports Generation

Analysts can generate beautifully formatted PDF reports for printing or offline presentation.

Company Tearsheets: Generates a 2-page document containing KPI grids, 10-year trend charts, balance sheet composition, and pros/cons text. Generated via: `python src/reports/batch_generate.py`

Sector Summary Reports: Generates summaries for each of the sectors listing constituents and their aggregate cash flow metrics. Generated via: `python src/reports/sector_report.py`

Portfolio Summary: General breakdown of the whole portfolio distribution. Generated via: `python src/reports/portfolio_summary.py`

7. REST API Reference (FastAPI)

The FastAPI REST server starts on port 8000. It exposes 16 endpoints for integration.

GET /api/v1/health: Health check, counts, uptime, version.

GET /api/v1/companies: List of all companies with parameters for sector, market cap, and search.

GET /api/v1/companies/{ticker}: Get full company profile and latest KPIs.

GET /api/v1/companies/{ticker}/pl: P&L; history array with custom year filters.

GET /api/v1/companies/{ticker}/bs: Balance sheet history array with year filters.

GET /api/v1/companies/{ticker}/cashflow: Cash flow history array with year filters.

GET /api/v1/companies/{ticker}/ratios: Pre-computed financial ratios.

GET /api/v1/companies/{ticker}/tearsheet: Download Tearsheet PDF binary.

GET /api/v1/screener: Screener endpoint matching min_roe, max_de, min_fcf, etc.

GET /api/v1/sectors: constituent counts and medians.

GET /api/v1/peers/{group_name}: Peer group constituent ranks.

GET /api/v1/companies/{ticker}/peers/compare: Radar chart metrics comparing company to peers.

8. Example curl Commands & Troubleshooting

Retrieve company data:

```
curl -X GET "http://localhost:8000/api/v1/companies/TCS"
```

Screener search request:

```
curl -X GET "http://localhost:8000/api/v1/screener?min_roe=20&max_de=0.5"
```

Troubleshooting common issues:

Port Conflicts (Error 98): If FastAPI cannot bind to port 8000, find and terminate the active process running on that port, or change the port using --port query parameter.

Missing Tearsheet PDFs: Ensure you run the batch generator first to populate reports/tearsheets/ directory before requesting pdf downloads from the API.